close-in terrestrial planets comparatively easy and more distant Jovian planets comparatively difficult to detect.

Planet C, some 2.8 times more massive than the Earth, orbits the pulsar every 98 days at a distance of 0.47 astronomical units* (AU); Planet B, with about 3.4 Earth masses, has a 67-Earth-day year at 0.36 AU. A smaller world, Planet A, still closer to the star, with about 0.015 Earth masses, is at 0.19 AU. Crudely speaking, Planet B is roughly at the distance of Mercury from our Sun; Planet C is midway between the distances of Mercury and Venus; and interior to both of them is Planet A, roughly the mass of the Moon at about half Mercury's distance from our Sun. Whether these planets are the remnants of an earlier planetary system that somehow survived the supernova explosion that produced the pulsar, or whether they formed from the resulting circumstellar accretion disk subsequent to the supernova explosion, we do not know. But in either case, we have now learned that there are other Earths.

The energy put out by B1257+12 is about 4.7 times that of the Sun. But, unlike the Sun, most of this is not in visible light, but in a fierce hurricane of electrically charged particles. Suppose that these particles impinge on the planets and heat them. Then, even a planet at 1 AU would have a surface around 280 Celsius degrees above the normal boiling point of water, greater than the temperature of Venus.

These dark and broiling planets do not seem hospitable for life. But there may be others, farther from B1257+12, that are. (Hints of at least one cooler, outer world in the B1257+12 system exist.) Of course, we don't even know that such worlds would retain their atmospheres; perhaps any atmospheres were stripped away in the supernova explosion, if they date back that far. But we do seem to be detecting a recognizable planetary system. Many more are likely to become known in coming decades, around ordinary Sun-like stars as well as white dwarfs, pulsars, and other end states of stellar evolution.

Eventually, we will have a list of planetary systems—each perhaps with terrestrials and Jovians and maybe new classes of planets. We